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Department of Computer Science and Engineering

Introduction to Artificial Intelligence and Machine Learning (CSN11601)

Lab Manual: Assignment-3

Lab Title: Programs on Decision and Loop Control Statements

Objective:

1. To understand and implement decision-making constructs like if-else and ternary operators in Python.
2. To practice loop control statements such as for loops and while loops.
3. To solve mathematical problems using Python, including finding the largest of numbers, computing GCD, and checking conditions using loops and conditional statements.
4. To apply the Pythagorean theorem in determining the type of triangle based on its sides.
5. To learn about special number classifications, such as Armstrong numbers, and generate sequences like the Fibonacci series.

Materials Required:

- Computer with internet access
- Python (latest version)
- Python Integrated Development Environment (IDLE)
- Access to an online Python environment (e.g., Google Colaboratory)

Theory:

Decision and loop control statements are essential tools in programming that allow the flow of a program to change based on conditions and repetitions.

Decision-making statements include:

- **If-else statements:** These statements allow the program to execute certain code blocks based on whether a condition is true or false.
- **Ternary operators:** A shorthand version of if-else statements that returns a value based on a condition in a single line.

Loop control statements include:

- **For loops:** Used for iterating over a sequence (such as a list, tuple, string, or range) with a defined number of iterations.
- **While loops:** Continue executing a block of code as long as a specified condition remains true, which is useful when the number of iterations is not known beforehand.

Mathematical computations:

- **Finding the largest of three numbers:** Utilizes comparison operators to determine the greatest value.
- **GCD (Greatest Common Divisor):** A mathematical function that finds the largest integer that can divide two numbers without leaving a remainder, commonly implemented using loops or recursion.
- **Pythagorean theorem check:** Involves verifying whether a set of three side lengths can form a right-angled triangle by checking the relationship $a^2+b^2=c^2$
- **Armstrong numbers:** Numbers that are equal to the sum of their own digits raised to the power of the number of digits. Checking for Armstrong numbers involves looping through the digits and computing powers.
- **Fibonacci series:** A sequence where each number is the sum of the two preceding ones, starting from 0 and 1. It's commonly implemented using loops or recursion.

List of Programs in Assignment-3

1) Write a python program to find largest of three numbers.
2) Read two integer numbers from the keyboard and print minimum value using ternary operator.
3) Write a Python program to compute the GCD of two numbers.
4) Write a python program to that accepts length of three sides of a triangle as inputs. The program should indicate whether or not the triangle is a right-angled triangle (use Pythagorean theorem).
5) Write a program to print a given number is Armstrong or not.
6) Write a program to print first 20 elements of a Fibonacci series